

Tatvik Sinha

Vellore Institute of Technology, Bhopal Campus
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Education

B.Tech in Computer Science & Engineering (Specialization in Artificial Intelligence and Machine Learning) Vellore Institute of Technology, Bhopal Campus	CGPA: 9.12/10 2024 – 2028
Class XII (Central Board of Secondary Education)	84.8%
Class X (Indian Certificate of Secondary Education)	92.3%

Technical Skills

Programming Languages: Python, Java, JavaScript, SQL
Data Science & ML: Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, XGBoost, Optuna, Jupyter Notebooks, Feature Engineering, Predictive Modeling
Deep Learning & AI: TensorFlow, PyTorch, NLP, spaCy, LLMs, Prompt Engineering
MERN Stack & Web: MongoDB, Express.js, React.js, Node.js, REST APIs, PostgreSQL
Cloud & DevOps: AWS, Docker, Git, GitHub, GitHub Actions, Linux

Experience

Machine Learning Intern, FlyRank AI	July 2026 – Present
- Built end-to-end predictive ML models (Python, Scikit-Learn) on large-scale datasets, performing exploratory data analysis and feature engineering to drive data-informed business decisions. - Developed and deployed NLP and automated analytics pipelines, leveraging deep learning techniques and cloud infrastructure (AWS) to extract actionable insights from unstructured data. - Engineered robust data preprocessing workflows – handling missing values, outlier detection, and feature scaling – improving model accuracy and reducing data pipeline latency by 30%.	
Backend Developer Intern, Shelter Zone NGO	May 2026 – July 2026
- Designed and deployed scalable RESTful APIs using Node.js and Express.js, building the data layer that powered real-time analytics dashboards for NGO outreach programs. - Architected optimized PostgreSQL schemas and wrote complex SQL queries for data aggregation, reporting, and analysis across 5+ interconnected service modules. - Implemented JWT-based authentication and role-based access control, ensuring data privacy and security for sensitive beneficiary datasets.	

Projects

BiFin – Enterprise Banking Intelligence Platform	<i>React, FastAPI, Python, Scikit-Learn, Optuna, Docker</i>
- Engineered a full-stack data science application (FastAPI + React) to predict customer churn from transactional banking data and recommend targeted retention strategies. - Built an end-to-end ETL and data ingestion pipeline in Python to clean, transform, and handle missing values across 50K+ transaction records, ensuring data integrity for downstream ML models. - Designed an automated machine learning pipeline with Optuna hyperparameter tuning and SHAP-based model explainability, enabling transparent and interpretable predictions. - Containerized the full application using Docker Compose, orchestrating seamless communication between the React frontend and Python ML microservices. - Achieved 87.40% accuracy and 0.8715 ROC-AUC via leakage-free cross-validation, exceeding baseline benchmarks by 12%.	
Resa-AI – Scientific NLP & ML Extraction Pipeline	<i>Django, Python, spaCy, XGBoost, Scikit-Learn, Docker</i>
- Architected a scalable data pipeline for autonomous PDF ingestion, text extraction, and multi-class sentence classification using NLP and ensemble ML models. - Engineered a feature engineering pipeline integrating TF-IDF vectorization, n-gram extraction, and feature scaling in Python (Pandas, NumPy) for high-dimensional text data. - Automated dataset acquisition and preprocessing from HuggingFace, performing extensive EDA and data quality analysis to identify and resolve class imbalance. - Built a comprehensive test suite with Pytest and integrated CI/CD via GitHub Actions, ensuring reproducibility and reliability of the data pipeline. - Evaluated Random Forest, XGBoost, and SVM classifiers, achieving 83.27% accuracy and 0.9278 ROC-AUC on 2,400+ sentences, deployed via Docker.	

Achievements

- Awarded the **NPTEL Gold Medal** in Cloud Computing by IIT Kharagpur, ranking among the **top 2%** of learners globally.
- Selected among the **top 15,000 candidates** nationwide in the prestigious Google BigCode competition.